ENGINEERING CHANGE NOTE (ECN)

PROJECT: PROJECT-CE updating to PROJECT-MYMILLI
ORIGINATION: Prof. Kidd, Instructor
PURPOSE OF CHANGE: Original Millis() function
DOCUMENT NUMBER: ECN 2.0
DATE OF PUBLICATION: 10/14/2026

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EXECUTIVE SUMMARY:

This engineering change notice documents the architectural changes made to project-CE to create project-MYMILLI. A header file called myMilli.h was created, which has the sole purpose of replacing the implementation of the Arduino native function millis(), while ensuring the same functionality. This was completed by utilizing the onboard timer present within the Ardunio's ATmega32, which was accessed through registers, counted, and when the timer completed, it reset and created an interrupt to flag that a millisecond has passed. All customer and high-level requirements (C.1-C.6, HL.1-HL.6) have remained unchanged and are preserved through reference. Tests have changed marginally. Tests T.1-T.8 remain the same same though test

T.9 has been added to ensure that the original myMilli function has the

exact same timing as the native millis() Arduino function.

HIGH-LEVEL DESCRIPTION:

Created a new .h file which replicates the functionality of the native Arduino millis() function using the ATmega32 built-in timer, registers, and interrupts.

REASON FOR CHANGE:

This change allows for a more reliable and flexible design, giving us more precise control that we can change in the future. Even though Arduino is open source, this change may save us from possible IP issues in the future.

Changed Documents:

No previous documents were changed; all new documents replaced existing ones

Added Documents:

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myMilli.h

Removed Documents:

No documents were removed

Replaced Documents:

Proj2.ino was replaced with Proj3.ino: The document is relatively the same, with the only changes being the required changes to properly set up and call the new functions, as well as adding the header file.

Any Additional Design Considerations:
No additional design considerations